Spatial Patterns of Hurricane Katrina Damage in New Orleans:

Socio-Economic Vulnerability, Racial Disparity, and Environmental Risk

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Midterm Report

SECTION 1

Written Analysis

Part 7 – Q1: Spatial Patterns of Hurricane Katrina Damage

The spatial distribution of damaged properties following Hurricane Katrina reveals clear geographic patterns across New Orleans that are closely related to socio-economic conditions, racial composition, and environmental vulnerability. By examining the maps produced in this analysis, it shows that storm damage was not evenly distributed across the city but instead concentrated in specific neighborhoods.

Map 1 shows the locations of geocoded damaged properties across New Orleans. The damaged points appear clustered primarily in the central and eastern areas of the city, suggesting that certain neighborhoods experienced more concentrated impacts from the storm.

When demographic variables are introduced, the spatial relationships become more significnt. Map 2, displays the percent Black population by census tract with damaged properties overlaid, shows that many damaged properties occur within tracts with higher proportions of Black residents. In contrast, Map 3, which maps the percent White population, indicates that many tracts with higher White population tend to have fewer damaged properties. These spatial patterns are supported by the results in Table 2, showing that 54.8% of damaged properties occur in census tracts above the median Black population percentage, and almost 70% occur in tracts where the Black population exceeds 25%. These findings suggest that predominantly Black neighborhoods experienced a disproportionate share of the storm's damage.

Moving on to the socio-economic patterns where map 4, compares damaged properties with median household income categories, shows that many damaged properties are located in census tracts with median incomes below $35,000. The summary statistics in Table 1 indicate that 51.2% of damaged properties occur in lower-income tracts, compared with 34.8% in higher-income tracts. This suggests that lower-income neighborhoods were more vulnerable to hurricane impacts, possible reasonings include poor housing conditions/ quality and worse flood protection. Additional spatial patterns appear in Map 5, which maps housing vacancy rates across the city. Areas with higher vacancy rates tend to overlap with clusters of damaged properties, showing that neighborhoods experiencing housing instability may have been susceptible to storm impacts and face greater challenges in recovery.

Environmental factors also play a key role in the distribution of damage. Map 7 highlights census tracts with the highest building damage rates, showing that the most severely affected tracts are concentrated in specific parts of the city. Meanwhile, Map 8 shows that many buildings are located in low-elevation areas, which are more vulnerable to flooding. This vulnerability is further illustrated in Map 9, where the lowest elevation zones are buffered to represent hazard exposure areas. Finally, Map 10 demonstrates that many buildings fall within or near these low-elevation hazard buffers, indicating that geographic exposure significantly influenced where structural damage occurred.

Overall, the spatial analysis reveals that Hurricane Katrina's impacts were shaped by the intersection of environmental risk and social inequality. Census tracts characterized by lower income levels, higher proportions of Black residents, and lower elevations experienced disproportionately greater levels of damage. This midterm analysis highlights the importance of integrating socio-economic conditions and environmental risk into urban policy and disaster prevention.

Part 7 – Q2: Broader Implications for Disaster Vulnerability and Urban Resilience

The spatial patterns observed in this analysis highlight how Hurricane Katrina's impacts were not evenly distributed across New Orleans, affecting low-income and predominantly Black communities. Lower-income neighborhoods often face structural disadvantages that increase their exposure to environmental hazards. Housing in these areas may be older, less well-maintained, or built with fewer resources dedicated to flood mitigation. Therefore, when extreme natural events happen (in this case Hurricane Katrina), these communities may experience greater levels of property damage. The analysis showing a higher proportion of damaged properties in census tracts with median household incomes below $35,000 reinforces the idea that economic vulnerability is closely linked to disaster impact.

Similarly, the racial patterns observed in the maps reflect historical and structural inequalities within the city. Many predominantly Black neighborhoods in New Orleans are located in lower-elevation areas that are more susceptible to flooding. These living patterns are the result of historical segregation and discriminatory housing policies over time. When Hurricane Katrina struck, these geographic and social conditions combined to produce disproportionately severe impacts in communities that were already economically and socially marginalized.

The elevation analysis demonstrates how environmental vulnerability interacts with social inequality. Areas located at lower elevations experienced higher building damage rates and greater exposure to flood hazards. Since many lower-income and predominantly Black neighborhoods are located in these low-elevated areas, environmental risk becomes closely tied to social inequality. This shows how disasters can amplify existing disparities where not all communities are affected equally.

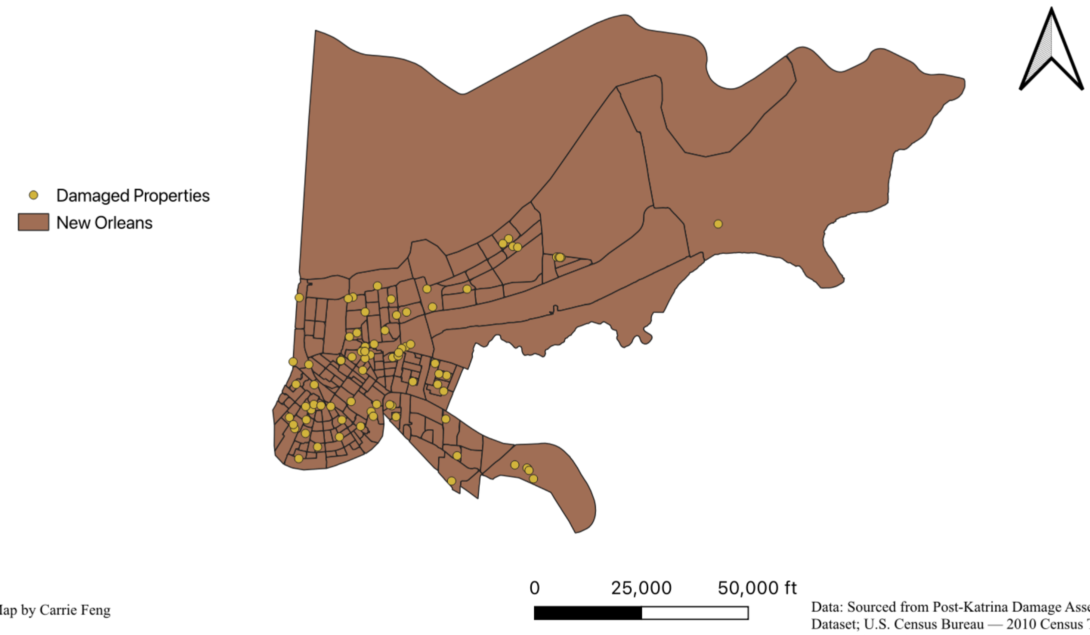
These findings have important implications for future urban planning and disaster recovery in New Orleans. First, flood mitigation and infrastructure improvements should prioritize the most vulnerable neighborhoods instead of focusing heavily on the well-off communities. Investments in better drainage infrastructure and flood-resistant housing can help reduce the disproportionate risks faced by low-elevation communities. Second, disaster recovery policies should focus on equitable rebuilding as a long-term resilience. Programs designed to support rebuilding and housing repair should make sure that low-income residents are not displaced and that recovery resources reach those most affected by disasters. Finally, urban planners and policymakers should consider demographic vulnerability, housing conditions, and environmental exposure when designing resilience strategies to build long-term resilience in communities like this. By addressing both environmental and social dimensions of risk, New Orleans can better prepare for future climate-related disasters and reduce the unequal impacts that events like Hurricane Katrina have caused.

SECTION 2

Maps 1 – 10

Map 1 – Part 2, Q1: Geocoded Damaged Properties

Map 1: Locations of Damaged Properties in New Orleans after Hurricane Katrina

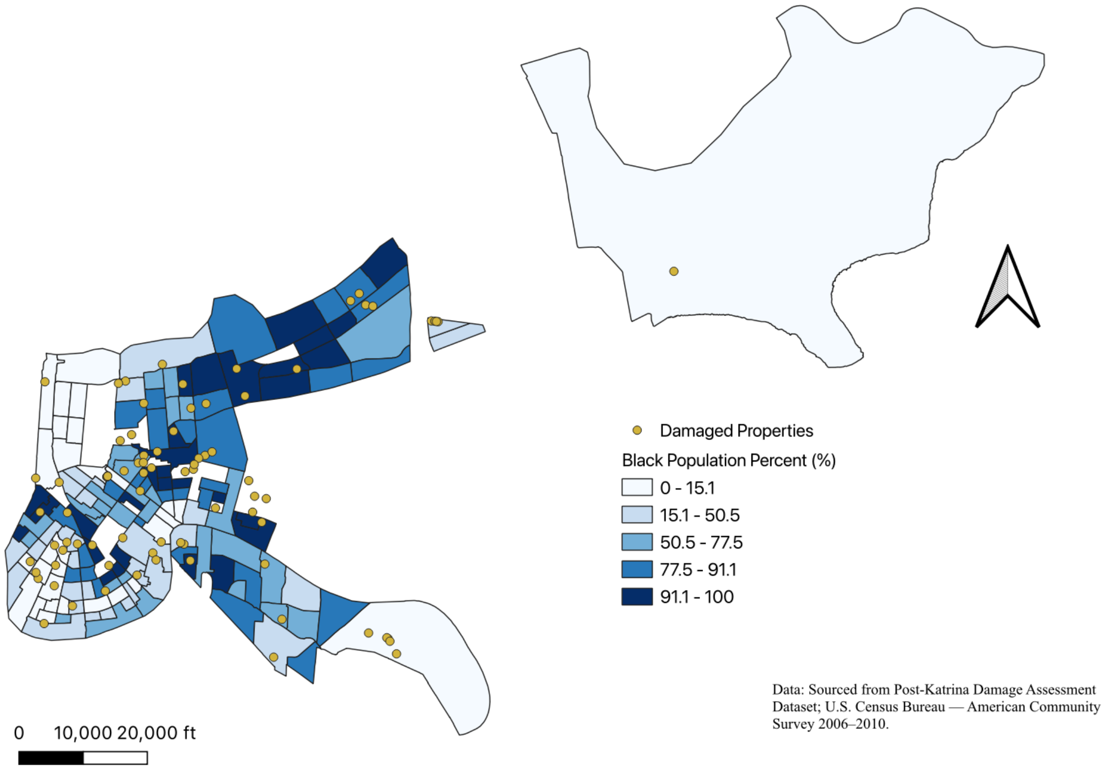


Map by Carrie Feng

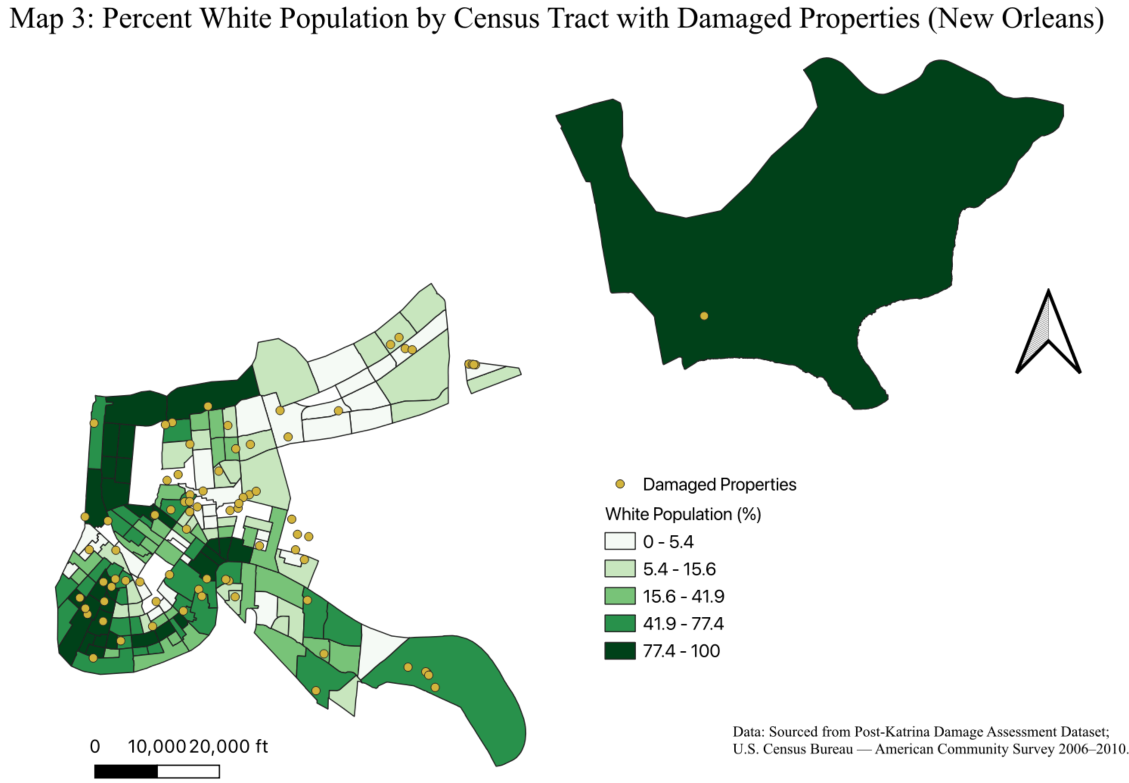
Data: Sourced from Post-Katrina Damage Assessment Dataset; U.S. Census Bureau — 2010 Census Tract Boundaries (Orleans Parish)

Map 2 – Part 2, Q2: Geocoded Damaged Properties – Black Population

Map 2: Percent Black Population by Census Tract with Damaged Properties (New Orleans)

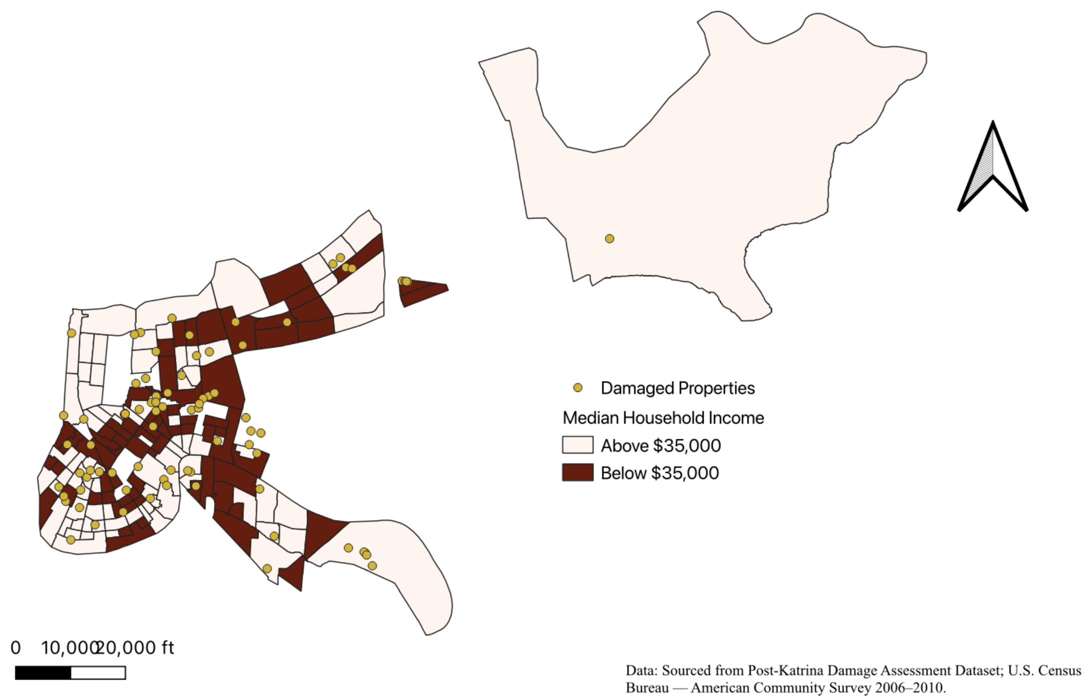


Map 3 – Part 2, Q2: Geocoded Damaged Properties - White Population



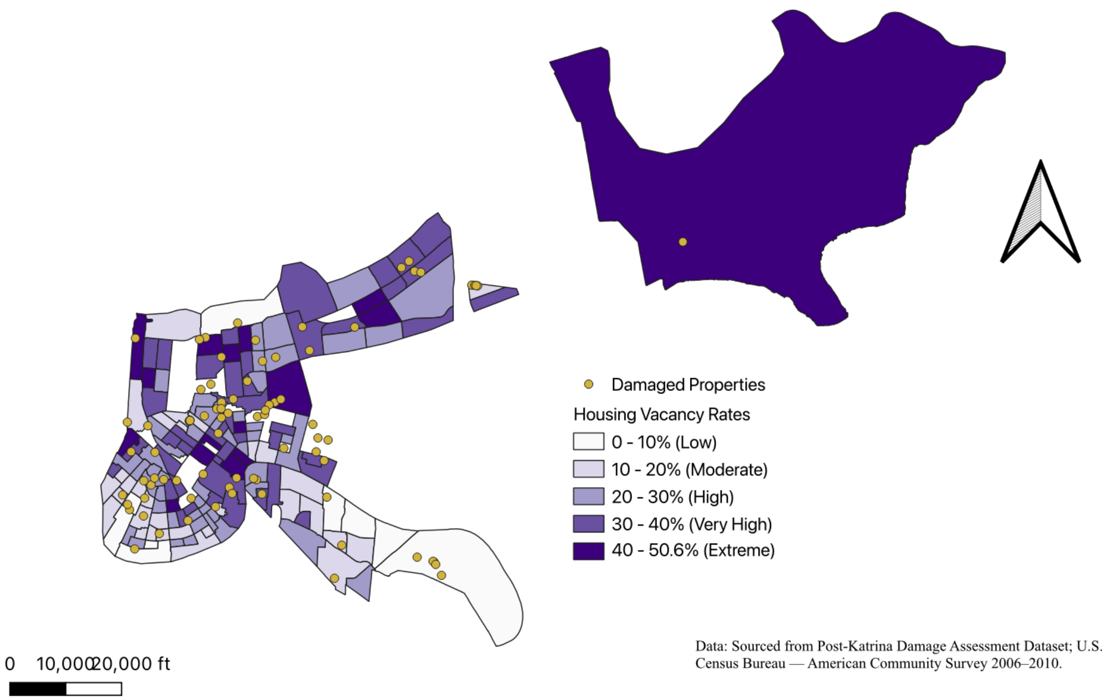
Map 4 – Part 2, Q3: Damaged Properties and Median Income

Map 4: Damaged Properties and Median Income by Census Tract (Above and Below $35,000)

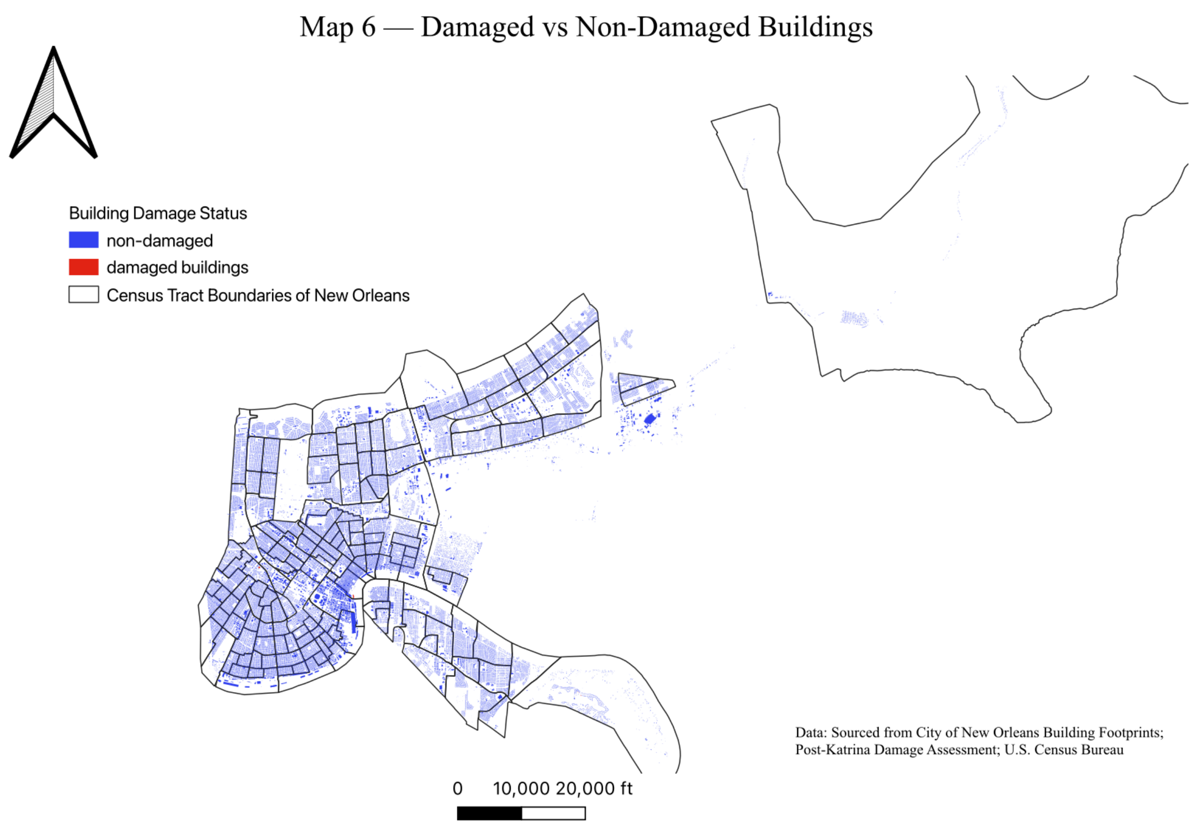


Map 5 – Part 2, Q4: Housing Vacancy Rates and Damaged Properties

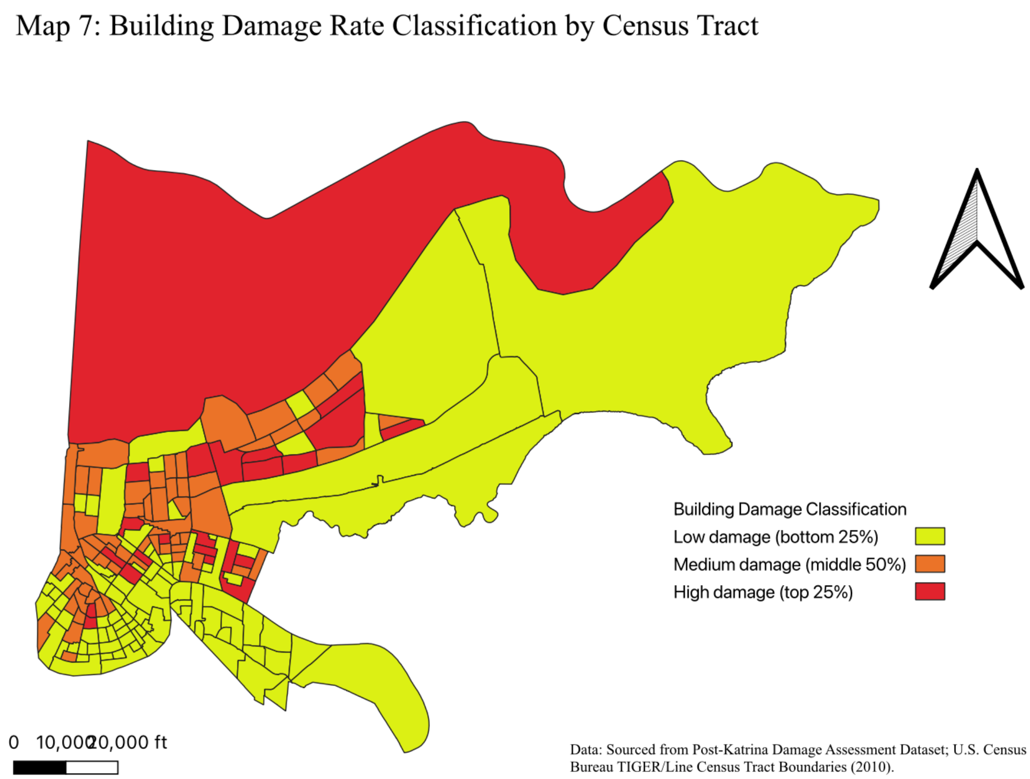
Map 5: Housing Vacancy Rates and Damaged Properties in New Orleans



Map 6 – Part 4, Q3: Damaged VS Non-damaged Buildings



Map 7 – Part 4, Q5: Building Damage Rate Classification by Census Tract

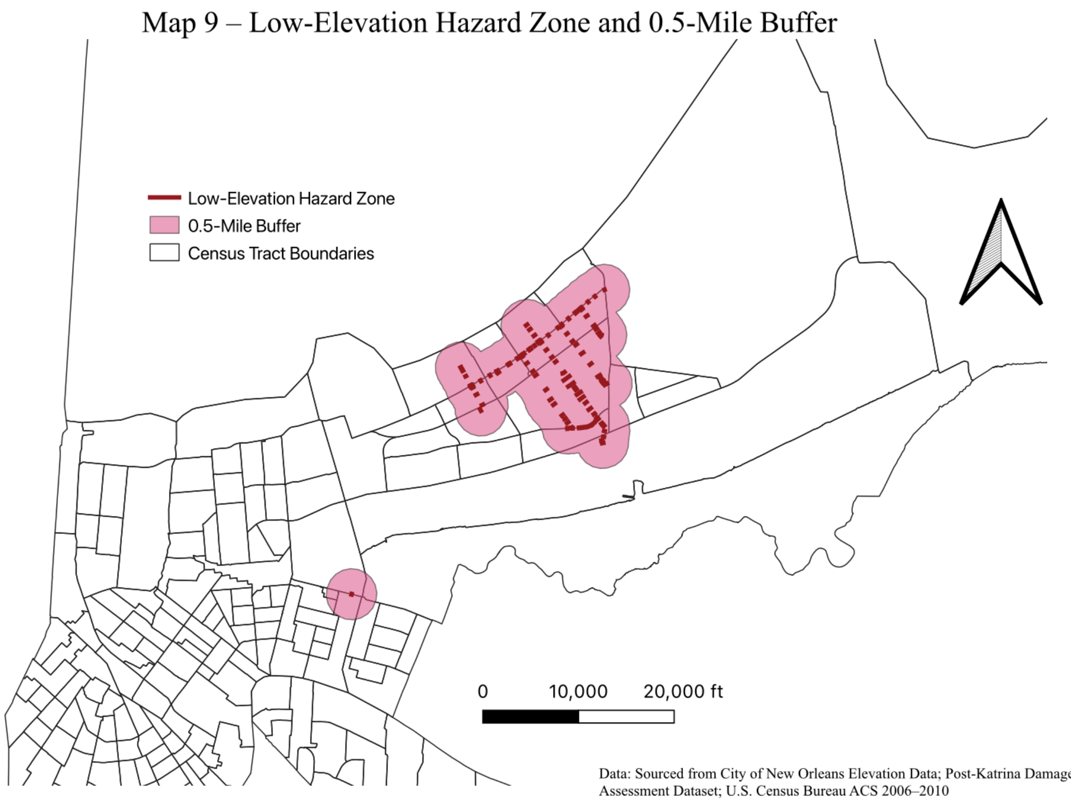


Map 8 – Part 6, Q2: Elevation Categories

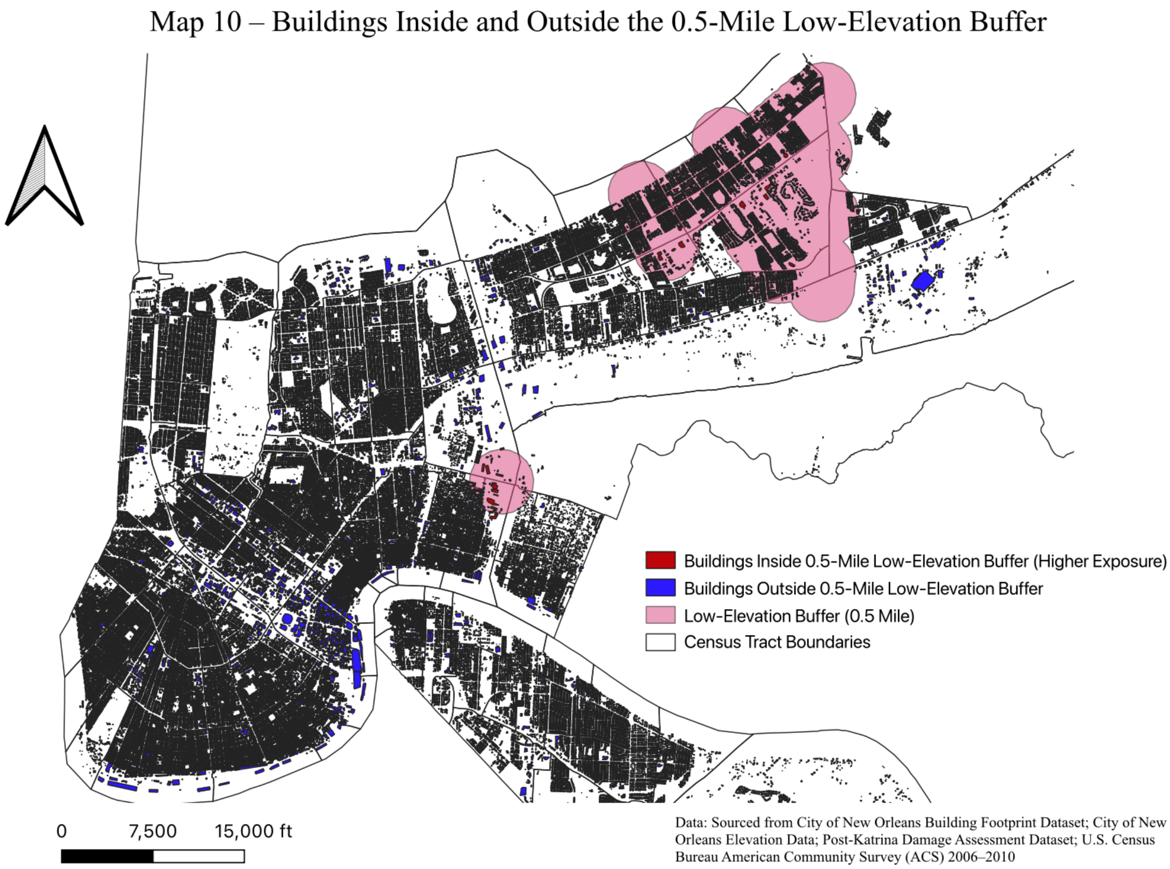
Map 8 – Elevation Categories in New Orleans

Data: Sourced from City of New Orleans Elevation Data; Post-Katrina Damage Assessment Dataset; U.S. Census Bureau (ACS 2006–2010)

Map 9 – Part 6, Q4: Low-Elevation Hazard Zone and Buffer



Map 10 – Part 6, Q5: Buildings Inside and Outside the Low-Elevation Buffer



SECTION 3

Tables 1 – 9

Table 1 – Part 3, Q1: Property Damage by Income Level

Income Level	Damaged properties (count)	Percent of Damaged Properties
Below $35,000	128	51.2%
At or Above $35,000	87	34.8%
NULL	35	14.0%
Total	250	100%

Table 2 – Part 3, Q2: Relationship on Household Income and Damaged Properties

Analysis	Category	Damaged Properties (count)	Percent of damaged properties
Median % black	Below median % Black	78	31.2%
Median % black	Above median % Black	137	54.8%
25% threshold	Below 25% Black	41	16.4%
25% threshold	Above 25% Black	174	69.6%

Table 3 – Part 4, Q1: Total Number of Damaged Points, Number Matched to Buildings, and Number Not Matched to Buildings

Metric	Value
Total damaged points	250
Matched to buildings	48
Not matched to buildings	202

Table 4 – Part 4, Q2: Building-Level Damage and Tract Dataset

table4

globalid	damage_count	damaged_building	GEOID10	building_area
{CA548704-E4D7-413C-AF25-2289246A409B}	0	0	22071001500	150.405092555098
{E924CFC0-F8AD-4F74-A529-E8473E1BB1E4}	0	0	22071001500	17.6688905013725
{6B903CBA-5C5B-4DC8-BDBD-F35E04AE15F0}	0	0	22071001500	96.0305151138455
{6DD6223C-4222-49B4-88BB-AE8DEED47AC7}	0	0	22071001500	132.161629206501
{B6028DB8-AC33-48AA-8ECC-7597D0635354}	0	0	22071001500	22.2577995089814
{753B4A79-EDF5-4B50-B686-ACCC8EAE18C5}	0	0	22071001500	168.796660110354
{457C1026-54D7-4345-90A7-EF6A59BEEC0E}	0	0	22071001500	224.091672642622
{59231456-B27F-4CA9-A16E-376F2680A64D}	0	0	22071001500	70.7102712923661
{85DF1B7F-706B-4BCE-9762-F812098D662C}	0	0	22071001500	44.1338497959077
{CEC249ED-DC16-4A82-8069-CF372501F0A5}	0	0	22071001500	168.108982175589
{EB7D8A73-B26A-4530-9752-A6C1A9405842}	0	0	22071001500	63.6702012224123
{479C2AD3-66C9-40DC-A8BC-5098042FAE17}	0	0	22071001500	182.883981458086
{75B9F987-7C7D-49EF-BC8C-BFEB9174F03E}	0	0	22071001500	109.211977744475
{B4620719-3516-445C-8869-023046A8690E}	0	0	22071001500	144.037037661998
{D44AC8E5-49F7-4ECD-9217-327DE90B04E2}	0	0	22071001500	150.427830013447
{00737BE3-AEF4-49D1-9BA6-3B1BB8B9D0D1}	0	0	22071001500	99.7083045886829
{09CEC33D-AEC3-46BE-B799-5F41F10F8113}	0	0	22071001500	139.494379877113
{A6CB7262-98A2-4977-A21E-37C202940BD8}	0	0	22071001500	136.428088149056
{8EFC4D6E-F747-4829-B6A7-F4068ECF25BF}	0	0	22071001500	332.461384834489
{453CA0B4-8CEB-4C42-A038-1CA381BC6E74}	0	0	22071001500	61.0888446066529
{EE23017F-A5B4-440E-B0C9-5B1683359FF5}	0	0	22071001500	129.481379517354
{181444DE-03C5-433A-8A97-3BD8D1F416F0}	0	0	22071001500	72.8749655745924
{B605C4D8-E246-4E88-92BA-936EADF26189}	0	0	22071001500	104.722385940171
{24DEEE64-8768-491E-A383-C3BD265D2609}	0	0	22071001500	16.6678524888121
{24B35C3D-CFFB-4DE1-B36A-C99DC327082B}	0	0	22071001500	116.727141675539

First 25 rows of the csv file, for the rest of the file, please refer to the csv file.

Table 5 – Part 4, Q3: Building Damage by Census Tract

FINALtable5

GEOID10	total_buildings	damaged_buildings	damage_rate
22071001500	645	0	0.000
22071012101	816	1	0.001
22071000604	1497	0	0.000
22071002100	506	1	0.002
22071008300	475	0	0.000
22071980000	181	0	0.000
22071012300	869	2	0.002
22071001747	1982	5	0.003
22071003100	627	1	0.002
22071001720	1650	5	0.003
22071001740	1442	6	0.004
22071007502	1277	2	0.002
22071000701	583	5	0.009
22071011600	737	0	0.000
22071009400	553	1	0.002
22071002900	562	1	0.002
22071010100	752	0	0.000
22071008500	437	0	0.000
22071006900	265	0	0.000
22071013200	1232	0	0.000
22071003900	537	0	0.000
22071001725	2442	3	0.001
22071007000	378	0	0.000
22071000617	538	0	0.000
22071002200	599	1	0.002

First 25 rows of the csv file, for the rest of the file, please refer to the csv file.

Table 6 – Part 5, Q1: Socio-Economic Characteristics of Top 5 High-Damage Tracts Compared to Citywide Distribution

Variables	Median Household Income ($)	Black Population (%)	Housing Vacancy Rate
Citywide Mean	41227.9636	60.6841	26.7127
Citywide Median	34464	73.134	26.962
Citywide SD	23893.2708	34.5410	10.6426
22071000701	30707	86.365	38.308
22071001735	32976	100	31.007
22071005000	22891	56.539	39.936
22071001401	26667	N/A	N/A
22071001740	24000	17.902	31.939

Table 7 – Part 6, Q1: Number of Tracts Excluded by Each Rule and Final Number of Tracts Retained

Category	Number of Tracts
Excluded: population < 500	11
Excluded: building count = 0	1
Exposure tracts retained	166

Table 8 – Part 6, Q3: Building Damage Rate by Elevation Category

Elevation Category	Total Buildings	Damaged Buildings	Building Damage Rate
-9 to -7	10043	19	0.0019
-7 to -4	25337	52	0.0021
-4 to -3	22042	30	0.0014
-3 to 20	105023	82	0.0008

Table 9 – Part 6, Q4: Building Damage Rate Inside vs. Outside Low-Elevation Buffer

Buffer Location	Total Buildings	Damaged Buildings	Building Damage Rate
Inside 0.5 mile low-elevation buffer	12083	23	0.0019
Outside 0.5 mile low-elevation buffer	150361	160	0.00106